Light – Reflection and Refraction — MCQ Worksheet

Science • Chapter 9 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The angle of incidence is equal to the:

(A) Angle of reflection
(B) Angle of refraction
(C) Critical angle
(D) Angle of deviation

Q2. Focal length of a spherical mirror is related to its radius of curvature R as:

(A) f = 2R
(B) f = R
(C) f = R/2
(D) f = R/4

Q3. A concave mirror always gives a virtual, erect image when the object is placed:

(A) At infinity
(B) At C
(C) Beyond C
(D) Between F and pole

Q4. A convex mirror always forms an image that is:

(A) Real and inverted
(B) Virtual, erect, diminished
(C) Real and magnified
(D) Virtual and magnified

Q5. Convex mirrors are used as rear-view mirrors because they:

(A) Magnify the image
(B) Give a wide field of view
(C) Form real images
(D) Produce no parallax

Q6. Mirror formula is:

(A) 1/v - 1/u = 1/f
(B) 1/v + 1/u = 1/f
(C) v + u = f
(D) v - u = f

Q7. Magnification produced by a mirror is given by:

(A) m = u/v
(B) m = -v/u
(C) m = v/u
(D) m = u + v

Q8. In sign convention, distances measured against the direction of incident light are taken as:

(A) Positive
(B) Negative
(C) Zero
(D) Infinite

Q9. When light goes from a rarer to a denser medium, it bends:

(A) Towards the normal
(B) Away from the normal
(C) Parallel to the normal
(D) Does not bend

Q10. Snell's law: n = sin i / sin r is also called the law of:

(A) Reflection
(B) Refraction
(C) Dispersion
(D) Total internal reflection

Q11. Refractive index of a medium n = c/v depends on the:

(A) Angle of incidence
(B) Speed of light in the medium
(C) Mass of the medium
(D) Density only

Q12. A glass slab causes:

(A) No bending of light
(B) Lateral displacement only
(C) Total internal reflection
(D) Reflection only

Q13. A convex lens converges parallel rays to a:

(A) Virtual focus
(B) Real focus
(C) Both
(D) Neither

Q14. A concave lens always gives an image that is:

(A) Real and inverted
(B) Virtual, erect, diminished
(C) Magnified
(D) Same size

Q15. The lens formula is:

(A) $1/v + 1/u = 1/f$
(B) $1/v - 1/u = 1/f$
(C) $v + u = f$
(D) $v \times u = f$

Q16. The unit of power of a lens is:

(A) Metre
(B) Watt
(C) Dioptre
(D) Joule

Q17. The power of a convex lens is:

(A) Negative
(B) Positive
(C) Zero
(D) Infinite

Q18. Power of a lens whose focal length is 50 cm is:

(A) +0.5 D
(B) +2 D
(C) +5 D
(D) +50 D

Q19. A magnifying glass uses a:

(A) Concave lens
(B) Convex lens with object within F
(C) Plane mirror
(D) Concave mirror

Q20. When the object is placed at the centre of curvature of a concave mirror, the image is:

(A) Real, inverted, same size
(B) Virtual, erect, magnified
(C) Real, inverted, diminished
(D) Virtual, inverted, magnified

Q21. Magnification produced by a lens is:

(A) $m = -v/u$
(B) $m = v/u$
(C) $m = u/v$
(D) $m = u-v$

Q22. For a concave lens, the focal length is taken as:

(A) Positive
(B) Negative
(C) Zero
(D) Infinite

Q23. When the object is at infinity for a concave mirror, the image is formed:

(A) At F, real, inverted, highly diminished
(B) At C, real, magnified
(C) At pole
(D) Between F and pole

Q24. The two laws of reflection are valid for:

(A) Plane mirrors only
(B) Concave mirrors only
(C) Convex mirrors only
(D) All types of mirrors

Q25. A real image:

(A) Cannot be formed on a screen
(B) Can be formed on a screen
(C) Is always erect
(D) Is always magnified

— End of Worksheet —